Strategy Land

Richard Rumelt's Web Journal on Good and Bad Strategy in Business, Politics, and Economics

The Public Pension Mess

Visualizing the risk in public pension funds and the extent of under-funding

March 15, 2011 By Richard Rumelt

The public pension system in the US is a mess. State and local plans for government employees are underfunded to an extent that the typical taxpayer will need to ante-up about $3,000 per year over 15 years to meet *past* commitments. And that ignores unstoppable future commitments. This mess has arisen out of a combination of bad doctrine, corrupt politics, and bad governance. The bad doctrine treats "expected" returns as if they are guaranteed over a long enough time horizon. The bad politics has politicians buying votes and campaign support from government employees in return for giveaways of unfunded benefits. The bad governance has fund managers chasing yield to magnify their incomes.

The Arithmetic of Saving

The idea of a pension is simple. People should save while they are earning in order to have an income after they retire. The question is "how much saving is enough?"

Suppose that Abigail is 35 and earns $100 per year ($100 thousand if that makes you happier). She will work for 30 more years, retire at 65, and live 20 more years after that. She wants to save enough to have a post-retirement steady income of $70 per year. There is no inflation, no taxes, no pay raises, and the interest rate is zero. The answer to this very simple problem is obtained by working backwards to the present. To have an income of $70 per year for 20 post-retirement years, she needs to have saved $1,400. To save that much over 30 working years, she needs to put aside $47 per year, almost one-half of her income.

It is unduly conservative to assume a zero interest rate on savings. What assumptions would be sensible? One simple risk-free[1] benchmark is the current yield on 30-year TIPS (inflation-protected treasury bonds). Today it is 2.19%. This instrument allows you to invest $100 today and receive $192 back in 30 years in "real" inflation-adjusted money. To compute how much Abigail should save with this assumption involves staight-forward compound interest calculations. This should be part of the skillset of every literate citizen, but, unfortunately, in the US even most college graduates cannot solve this problem. Out of this general innumeracy grows layer upon layer of financial advisory services supported by a rat's nest of bad legislation and regulation.

To maintain an income of $70 per year for 20 years, and given an interest rate of 2.19%, Abigail would need to have saved a capital amount of $1,124 by age 65:

$$(70/.0219)\ (1- 1/1.0219^{20}) = 1124 \text{ (or, Excel function PV(0.0219,20,-70).)}$$

To accumulate that much, she would have to save $27 each working year, slightly more than one-fourth of her income.:

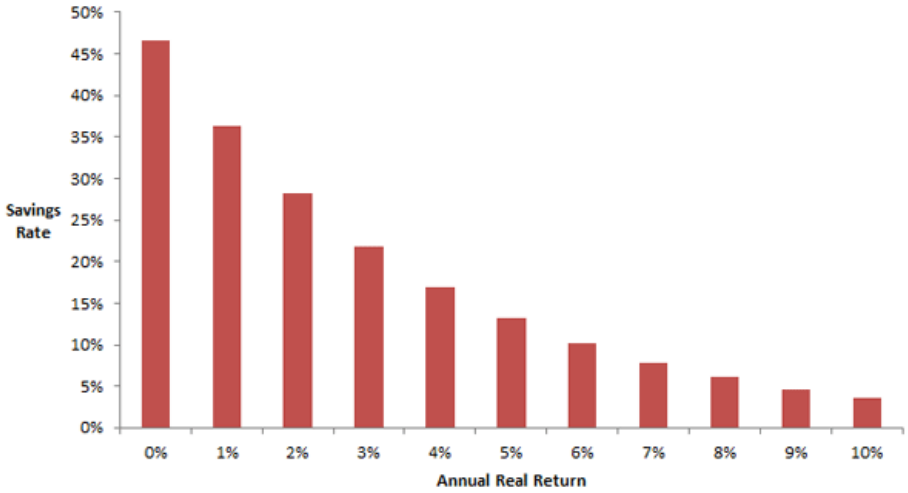
$$1124\ (0.0219)/(1 - 1/1.0219^{30}) = 27 \text{ (or, Excel function PMT(0.0219,30,-1124).)}$$

This rate of saving (27% of income) is about what Japanese workers used to save and is about the rate of household saving today in China. To modern Americans, it seems huge. To tuck away almost 30% of your income! How would you afford a new SUV every five years? Yet, there is no getting around the cold fact that this is about how much one would have to save to be ***certain*** that the necessary $1,124 would be there at age 65.

The current 30-year TIPS rate is lower than it has been. Back in 2001, it was closer to 3.5%. Were that the base assumption, Abigail would have to save less—19% rather than 27% each year, though still a large fraction. Unfortunately, the risk-free rate wobbles around over time. Today, the 30-year real risk-free rate is 2.19%. Tomorrow it may be higher or lower.

Most of us do not save 27% of our income. The reason our employee pension plans put away less than 27% of our wages is not financial magic. It is hope. There is a vast self-interested industry aimed at convincing people that it makes more sense to save less and take some risk. Thus, the typical public pension fund assumes an annual return of 8% counter balanced by an assumed inflation rate of 3%, giving a net real (expected) annual return of 5%. The rationale for the higher rate is that such rates have been earned, on average, in the past, by investing in stocks, real estate, and other volatile assets.

Changing the 2.19% interest rate assumption to 5% brings the required savings rate down to 13% of earnings. If the employer is matching, the hit seen on the paycheck is only 6.5%. As the chart below shows, the effect of the assumed rate of return on the savings rate is very powerful.



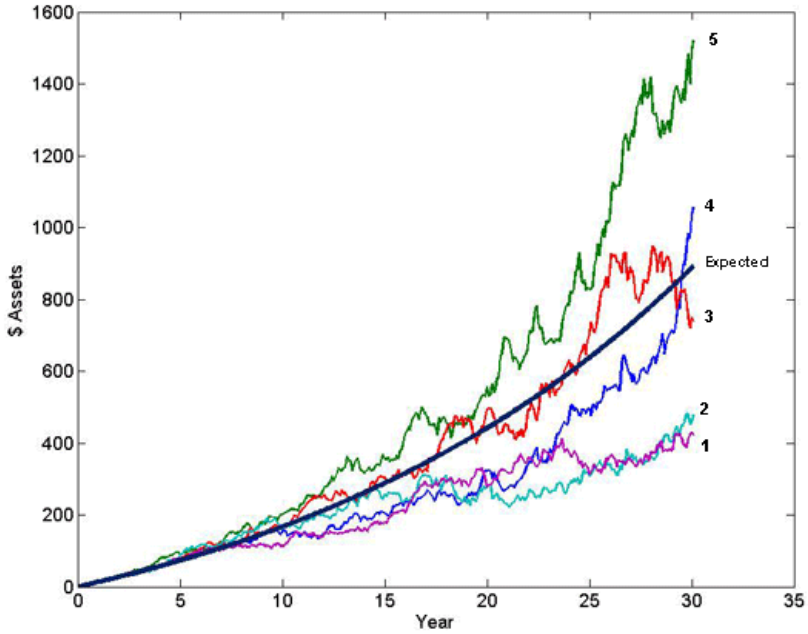
How reasonable is it for Abigail to bear the riskiness of an *expected* 5% real (8% nominal) annual rate of return? It seems like a small jump from 2.19% to 5%. Surely it is just a matter of making small adjustments along the way.

Visualizing The Risk

If Abigail saves $13 each year, or 13% of her income, then we expect that her portion of the fund—her nest egg—will grow 5% each year to reach $872 over 30 years. The heavy black line on the chart below shows the steady expected growth of her nest egg under the constant proportion savings plan.

However, the 5% annual return is not certain—it wobbles. In fact the standard deviation of annual real returns on stocks, over long periods of time, is about 18%.[2] The typical public employee pension fund portfolio is not pure stocks, and the standard deviation of its annual return will be more like 12%. That is, about two-thirds of the time the inflation adjusted annual rate of return will between -7% and +17%. To get a more realistic picture of what may happen I have computed five different equally likely scenarios.

In these simulations, Abigail puts $100/12 into her pension fund each month. And, each month the plan balance grows (ir shrinks) by a random amount. These random monthly returns are computed so that the expected annual return is log-normally distributed with mean 1.05 and standard deviation 0.12.[3] As you can see, the move from 2.19% with certainty to 5% with risk is very consequential.



Five Scenarios For Abigails-Nest-Egg

Looking as the chart, you can see that in three possible futures there is not enough capital to fund Abigail's expected pension. Under scenarios 1 and 2, and if the plan is a defined contribution plan, this means she might have to make due with a pension only one-half of size she hoped for. She will have forego the world tour, move back in with her children. and eat a (healthy) diet of beans and rice.

Why are two of the five futures so dismal? To some extent this is just the luck of the draw–it is the outcome ordained by the random numbers used to create the simulation. But look more closely at the chart and you will see a very positive scenario (green 5), where Abigaile's nest-egg grows to $1,500, about twice the target. On average, this kind of very positive outcome balances out the less attractive outcomes, producing an expected 5% annual rate of return. This is exactly what is meant by "taking more risk to get a higher ***expected*** return." Unfortunately, what you ***expect*** is not what you get. There is no financial "fix" for poor performance in the overall market for risky assets. This risk is only diversifiable across alternative universes.

Why did Abigail agree to take on this risk?

- Everyone else was doing it and Abigail is no whiz at finance or statistics. More importantly, the fund manager assured everyone that the risk was "reasonable."
- Abigail likes the idea of having to save less.
- If the fund manager were a perfect fiduciary, he would have shown Abigail and the other employees a chart like this. But, the fund manager probably has an incentive agreement with the fund's board giving him a very large bonus if the fund's short-term return beats the "market" or some peg (like 8%). Of course, if he fails to beat the market, there is no obligation for him to make up the losses out of his pocket. Hence the fund manager *prefers risk* and has an incentive to create a riskier investment program. (See "How Bonuses Promote Excessive Risk-Taking" for more about this.)
- The fund manager is probably about 45 years old and will be long-retired himself by the time Abigail turns 65.
- All the other fund managers are doing this.
- The employer is in competition with other employers. If one employer uses an honest fiduciary and others do not, then prospective hires will be attracted to the employer promising more take-home pay.

Over and Under Funded Plans

If the plan is a defined benefit plan, then Abigail will probably get her pension, but the plan will be underfunded. There will be a big battle between employees and various levels of government over who will make up the shortfall. Private pension plans are not allowed to get underfunded for any substantial length of time. But public pension funds are not held to the federal ERISA guidelines and are normally expected to be no more than 80% funded. The idea seems to be that taxpayers are the funders of last resort and will, in the end, make up the difference.

One problem with this set up is that the taxpayers are not granted a rebate if the market does well. If the pension plan gets overfunded (for example, scenarios 4 and 5 on the chart), politicians will use the surplus to increase employee wages and/or benefits. For example, during the 1996-99 stock dot-com bubble, the giant California public employees fund, CalPers, estimated that its assets were 138% of its liabilities. It proposed that the state and cities in California increase benefits. Like the bonus to the fund manager, this is a one-way ratchet on public funds. Heads the managers and employees win, tails the taxpayer loses.

To see the trouble this kind of behavior can cause, look at scenario 3 on the chart. In this case the pension plan became over-funded (above the expected line) in years 25-28. But poor performance subsequently dragged it back down into a under-funded state. Had the fund management taken the "excess" and given it away in year 27, the outcome by year 30 would have been much worse.

As another example, the University of California (my employer) has a separate pension plan for University employees. Right now its assets are only 71% of its liabilities. The shortfall is about $15 billion. The current (recently raised) contribution rate is about 8%, split 3/4ths by UC and 1/4 by the employee. To fully fund the plan over the next 15 years, 38% of compensation would have to be put into the plan. To deal with this huge amount of underfunding, the University plans to increase the contribution rate to 10.5% in July 2011 and 15% in 2012. In other words, the

"leaders" are slightly reducing the speed of the ship in the hopes that the looming iceberg will melt before the ship hits it. But some years back, in the mid-to-late 1980s the plan was well over-funded. What the University did then was create a temporary program giving super-early high-pension retirements to senior professors. The University was not allowed to tap the pension fund for operations or to pay salaries, but it got around those constraints by replacing a group of high-salary employees with lower-paid young hires. This maneuver lowered the annual wage bill for working employees. (Of course, total payments rose, but no one looks at total payments.) This sort of lopsided governance is the heart of the problem in public pension plans.

The situation is presently quite difficult for public pensions across the United States. In a careful study, Robert Novy-Marx and Joshua Rauh[4] have shown that local governments (city and county) across the nation have a total underfunding of $383 billion, or $14,000 per local household. This is in addition to the $3 trillion in underfunding of state level pension plans, or almost $27,000 per household. The table below shows the underfunding by local area (cities or counties) and the associated state situation. The worst situation is Chicago, with a city underfunding of $42,000 per household on top of an Illinois underfunding of $35,000 per household. This total underfunding per Chicago household of $76,000. In the final column of the table I have calculated annual taxes per household that would be needed to make up the present underfunding over 15 years. I have not computed the additional taxes that would be needed to counter continuing underfunding in the future.

Will Chicago taxpayers pony up the $7,410 per year? One doubts it. Rather, the city will ask the state to step in. And the state will ask the federal government to step in. Just because you live in Nebraska doesn't mean you are off the hook.

		Unfunded Pension Liabilities					Estimated Annual Tax To Fund
		$ Billions		$ per HH			
City/County	***State***	***Local***	***State***	***Local***	***State***	***Total***	***Total***
Chicago	Illinois	44.8	167.3	41,966	34,943	76,909	7,410
San Francisco	California	8.7	370.2	34,940	29,485	64,425	6,207
New York City	New York	122.2	132.9	38,886	18,498	57,384	5,529
Boston	Massachusetts	7.5	54.2	30,901	21,630	52,531	5,061
Cincinnati	Ohio	2.0	166.8	15,681	36,546	52,227	5,032
Hartford	Connecticut	0.7	49.0	14,333	36,404	50,737	4,888
Los Angeles	California	26.1	370.2	18,193	29,485	47,678	4,593
Nashville Davidson	Illinois	2.3	167.3	10,048	34,943	44,991	4,335
Alameda	Ohio	4.2	166.8	7,579	36,546	44,125	4,251
Stanislaus	Connecticut	1.3	49.0	6,698	36,404	43,102	4,153
Contra Costa	California	5.0	370.2	12,771	29,485	42,256	4,071
Santa Barbara	California	1.8	370.2	11,995	29,485	41,480	3,996
DeKalb	Ohio	1.4	166.8	4,873	36,546	41,419	3,990
San Jose	California	4.1	370.2	11,391	29,485	40,876	3,938
Milwaukee	Wisconsin	3.4	56.2	14,853	25,054	39,907	3,845
San Mateo	California	2.5	370.2	9,415	29,485	38,900	3,748
San Joaquin	California	2.3	370.2	9,119	29,485	38,604	3,719
Sacramento	California	4.5	370.2	8,582	29,485	38,067	3,667
Fresno	California	2.9	370.2	8,401	29,485	37,886	3,650
Orange	California	9.3	370.2	8,233	29,485	37,718	3,634
Philadelphia	Pennsylvania	9.7	100.3	16,690	20,436	37,126	3,577
Los Angeles	California	27.6	370.2	7,473	29,485	36,958	3,561
San Bernardino	California	5.1	370.2	6,716	29,485	36,201	3,488
Baltimore	Maryland	3.7	43.5	15,420	20,446	35,866	3,455
San Diego	California	7.2	370.2	6,329	29,485	35,814	3,450
Detroit	Michigan	6.4	63.6	18,643	16,366	35,009	3,373
Cook	California	8.2	370.2	4,112	29,485	33,597	3,237
Fresno City	California	0.7	370.2	3,647	29,485	33,132	3,192
Fulton	California	1.3	370.2	3,276	29,485	32,761	3,156
St Paul	Missouri	1.4	42.1	13,686	17,863	31,549	3,039
Kern	Missouri	3.6	42.1	11,919	17,863	29,782	2,869
Montgomery	Alabama	3.0	40.4	8,118	21,628	29,746	2,866
Dallas	Texas	6.3	142.3	12,856	16,456	29,312	2,824
Houston	Texas	9.1	142.3	10,804	16,456	27,260	2,626
Baltimore	Maryland	1.9	43.5	6,577	20,446	27,023	2,603
Fairfax	Indiana	5.6	30.2	14,415	12,134	26,549	2,558
Arlington	Texas	0.8	142.3	10,000	16,456	26,456	2,549
Seattle	Washington	2.1	42.9	9,125	16,846	25,971	2,502
Tacoma	Washington	0.7	42.9	9,082	16,846	25,928	2,498
Ventura	Washington	2.5	42.9	8,195	16,846	25,041	2,413
Jacksonville	Florida	4.0	89.8	12,994	12,046	25,040	2,412
Tulare	Arizona	0.7	48.7	4,068	20,163	24,231	2,334
Fort Worth	Texas	2.0	142.3	7,212	16,456	23,668	2,280
Anne Arundel	Michigan	1.4	63.6	7,081	16,366	23,447	2,259
Phoenix	Arizona	1.9	48.7	3,176	20,163	23,339	2,249
Sonoma	South Dakota	1.5	4.7	8,394	14,869	23,263	2,241
Miami	Florida	1.6	89.8	9,689	12,046	21,735	2,094
San Antonio	Texas	1.7	142.3	3,201	16,456	19,657	1,894
Memphis	Tennessee	2.1	23.2	8,432	9,283	17,715	1,707
Tampa	Florida	0.3	89.8	2,309	12,046	14,355	1,383

Sources:
Pension assets and pension liabilities: Calculations by Robert Novy-Marx and Joshua Rauh, as of June 2009
Public Pension Promises: How Big Are They and What Are They Worth? (Novy-Marx & Rauh), October 2010)
Tax estimates are 15 year amortization at 5% interest.(calculated by R. Rumelt)

Local & State Unfunded Pension Liabilities

(Click here to download the spreadsheet).

The main academic critique of the public pension system, including Novy-Marx and Rauh's work, centers on the method for measuring liabilities. The actuarial industry's method, enshrined in mistaken legislation, is to discount expected payouts at the expected market rate of return, say 8 percent. The problem is that the payouts are almost certain whereas the 8 percent is a return on risky assets. To correctly measure the present value (or cost) of almost certain future payouts one should use the debt yields, not stock market or portfolio rates. Discounting expected payouts at the much lower debt rates, as Novy-Marx and Rauh do, increases the liability figure, making the pension plan's status look worse.

Correcting this accounting mistake helps give a more accurate picture of the bungled system. It is, however, a very complex job for an outsider to undertake because the pension funds only report their own balances, not the patterns of future cash flows leading to them. Thus, an outsider trying to untangle the true liability situation must make estimates of future commitments to existing employees, to future new hires, of in-payments from employees, governments, and market fluctuations.

One huge step forward in transparency would be for public pension funds to provide best-estimates of the actual cash payouts they anticipate in each future year. Even more sensible, given that we work with computers and not quills and inkwells, would be for the funds to reveal the models they use to forecast those cash flows.

The overarching problem, however, is not simply bad accounting. It is bad doctrine combined with bad governance. As future entries will show, the bad doctrine concerns returns on financial assets and the bad governance arises from institutionalized myopia.

Footnotes

1. Some will argue that the price of a long-term bond, like a TIP, is quite volatile and hardly risk-less. However, if the program is to buy and hold for 30 years, there is no risk. The intermediate price fluctuations don't matter. [↩]

2. See John Y. Campbell, Peter A. Diamond, and John B. Shoven, "Estimating the Real Rate of Return on Stocks Over the Long Term," Social Security Advisory Board, 2001. http://www.ssab.gov/Publications/Financing/estimated%20rate%20of%20return.pdf [↩]

3. More technically, the differences between annual observations of the log portfolio value are normally distributed with mean .0423 and standard deviation 0.1139. The simulation was carried out in monthly increments wherein the normal distribution had mean .0424/12 and standard deviation .1139/sqrt(12). More complex models of stock price movement are available, but make little difference over a long 30-year term. In particular, assumptions which fatten the tails of the distribution tend to be canceled out by equally valid assumptions about mild return-to-the-mean behavior. [↩]

4. "The Crisis in Local Government Pensions in the United States," October 13, 2010. [↩]

Tagged finance, governance, management, pension fund | 3 Comments

3 Responses to *The Public Pension Mess*

Alan Lambert [March 17, 2011 at 8:20 am]
I like your approach to pricing the liability—so many dollars per household per year in extra taxes. If we priced Medicare and Social Security and our various wars this way, it might make people think twice.

Reply

Cate Delphia [March 20, 2011 at 10:23 pm]
You have not covered the stinking heart of the matter. Politicians run up these humungous debts by paying off unions who make political contributions and who get out the vote. We have sunk to banana republic status.

Reply

Scott Mosier [April 18, 2011 at 4:01 am]
The analysis is interesting (and scary) but the solution is complicated. I believe that more transparency is helpful but I am not sure that it is it the lack of transparency that has led us down this path. Even armed with this knowledge, what is likely to change? As you point out, bad governance and political incentives seem to be at the heart of the matter. It will be interesting to see if the current Tea Party/Republican agendas can make any progress on this issue. The attempts to de-certify the unions in Wisconsin and other states appears to be one way to reduce the incentives for politicians to continue to pander to unions with extravagant promises that make little financial sense and will only result in further taxpayer burdens in the future. I hope the public will eventually elect leaders that understand this basic math problem but I'm not holding my breath.

Reply

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